Question 1: Phosphate-buffer

Commercially manufactured buffer solutions have a relatively high price, especially if they are used in huge quantities in laboratories. The company BioGen assigns their analytical department to clarify the exact composition of the buffer most used in their biochemical department, in order to be able to produce this buffer on their own. The only information available from the commercial producer is a list with chemicals used to compose the buffer in question: phosphoric acid 85% (m/m) reagent grade, sodium dihydrogenphosphate dihydrate, di-sodium hydrogenphosphate – heptahydrate, tri-sodium phosphate – dodecahydrate.

The buffer has a pH-value of 6.50. The acid constants of phosphoric acid are $pK_{a1} = 2.23$ $pK_{a2}=7.21$, and $pK_{a3}=12.32$.

At first the total amount of phosphorous containing species is determined by a permanganometrical titration. Thereby the buffer is acidified slightly by adding acetic acid. Then ammonium chloride is added, and the phosphate is precipitated quantitatively with uranyl(VI)-acetate solution as insoluble uranyl ammonium phosphate ($NH_4UO_2PO_4$). In a next step the purified uranyl salt is reduced with aluminium plate in sulphuric acid solution to uranium(IV). In the end the uranium(IV) is titrated with potassium permanganate, where uranium(IV) is oxidised to produce UO_2^{2+} again.

1.1. Write down a balanced equation for the titration.

1.2. In this titration 10.00 mL of buffer solution consume 13.34 mL of potassium permanganate solution (0.0200 mol/L). Calculate the total concentration of phosphorous containing species.

1.3. Now calculate the individual concentrations of the phosphorous containing species in the buffer solution. Assume that at the given pH some of the P-containing particles are present only in very low quantities and therefore may be neglected. Using this assumption determine the concentrations of the remaining P-containing particles. Taking these values now calculate the concentrations of those P-containing particles which were neglected beforehand. Check the correctness of your assumption (the deviation should be less than 2%).

A main disadvantage of this buffer is the precipitation of calcium ions eventually present as apatite $Ca_3(PO_4)_2$. The solubility product of apatite is $1.00 \cdot 10^{-26}$.

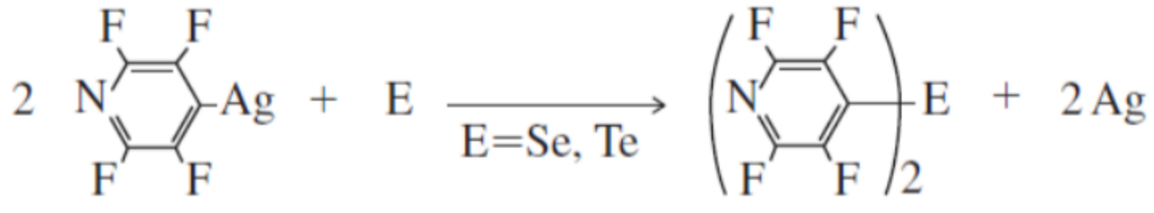
1.4. Calculate the possible maximum concentration of dissolved calcium ions present in this buffer.

1.5. 2.00 g of solid NaOH are added to one litre of this buffer solution. Calculate the pH-value of the resulting solution as well as the concentrations of all the other ions present applying useful approximations.

Question 2:

Elemental Se and Te react with 4-tetrafluoropyridyl silver(I) to afford $Se(C_5F_4N)_2$ and $Te(C_5F_4N)_2$. Two independent bent molecules were found for each compound in the solid state with C-Se-C angles of 95.47(12)° and 96.16(13)°, and C-Te-C angles of 90.86(18)° and 91.73(18)°, respectively (Aboulkacem, S.; Naumann, D.; Tyrra, W.;

Pantenburg, I. *Organometallics* , **2012** , *31* , 1559).



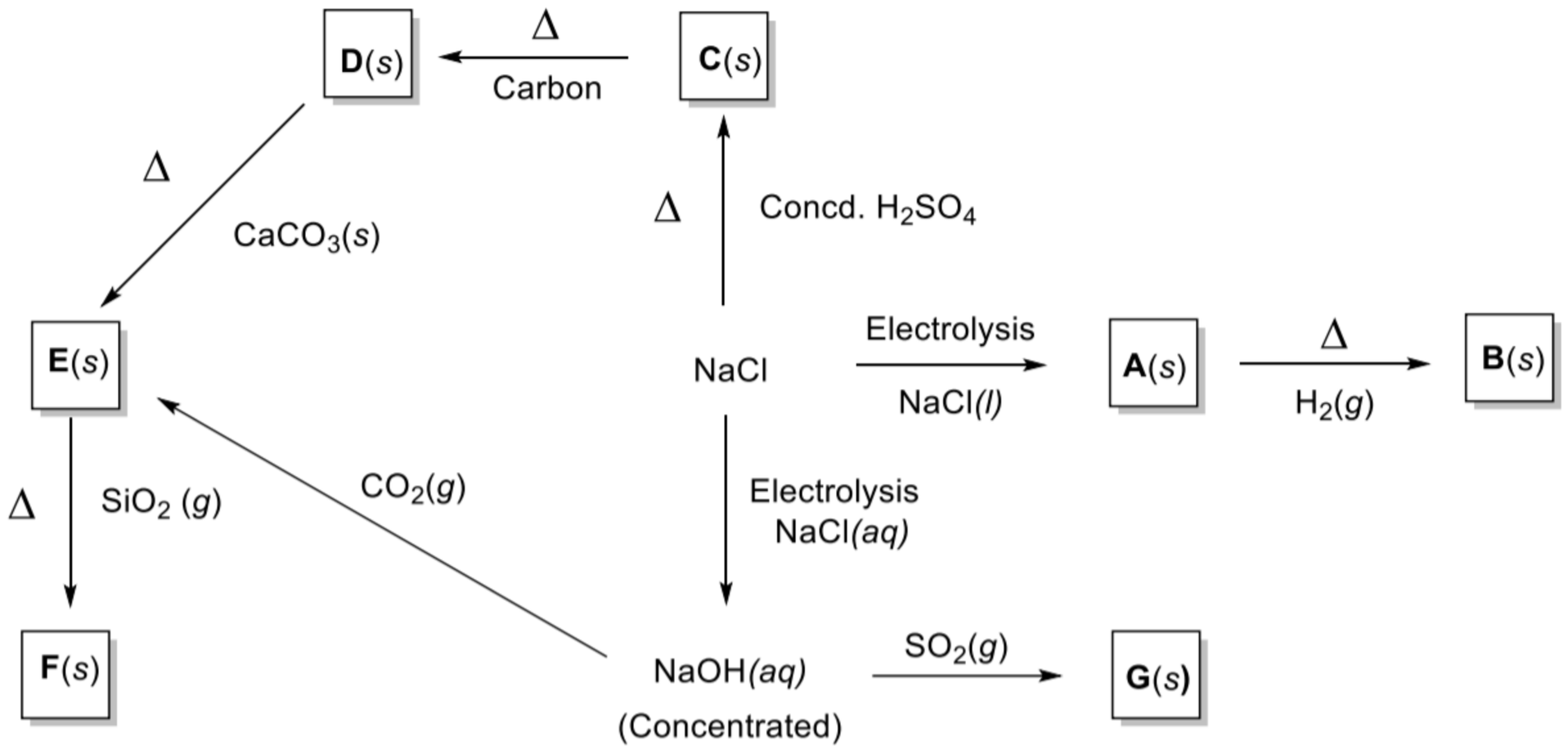
a. Explain why the angles are more acute for the Te compound relative to the Se compound.

b. These angles are approximately 0.8° (Se) and 2.0° (Te) more acute than those in the related pentafluorophenyl (C_6F_5) compounds. The greater compression of these angles in the 4-tetrafluoropyridyl compounds has been postulated on the basis of group electronegativity differences. Explain the logic associated with this hypothesis.

Question III: Sodium Compounds from Salt

The Salt Lake basin in Turkey is of great importance for the conservation of biological diversity and is classed as a wetland according to international criteria. It is also one of Turkey's richest lakes for the presence of birds. There are 85 bird species, 129 insect species (4 of which are endemic), 15 mammal species, and 38 endemic plant species. Some 40% of Turkey's salt needs (as table salt) are supplied from this lake. Salt in the Salt Lake is formed by meteorological waters draining underground and melting the previously formed salt domes and carrying them along the tectonic lines. Salt production in the Salt Lake is done by evaporation of lake water under the sun. A pooling system is used in the salt production with solar energy.

Table salt is one of the most common household chemicals. It is 97% to 99% sodium chloride, which is an ionic compound with the chemical formula NaCl, representing a 1:1 ratio of sodium and chloride ions. NaCl is the compound most responsible for the salinity of seawater and of the extracellular fluid of many multicellular organisms. In its edible form of table salt, it is commonly used as a condiment and food preservative. A second major application of sodium chloride is de-icing of roadways in subfreezing weather. Large quantities of sodium chloride are also used in many industrial processes such as the chloro-alkaline industry and soda-ash industry as well as in miscellaneous industrial uses: water softening, medicine, agriculture, firefighting, and cleanser. NaCl is used, directly or indirectly, in the production of many sodium compounds, which consume most of the world's production. The scheme below shows the preparation of some sodium compounds starting from NaCl.



Preparation of some sodium compounds starting from NaCl.

3.1. Write the formulas of products A–G.

Sodium carbonate (Na_2CO_3, soda ash) is used primarily in the manufacture of glass, which is produced mostly from natural sources, such as the mineral trona, $Na_2CO_3{\cdot}NaHCO_3{\cdot}nH_2O$. It can be also manufactured mostly from NaCl, $CaCO_3$, and NH_3 using a process introduced by the Belgian chemist Ernest Solvay in 1863. The key step involves the reaction of NH_3(g) and CO_2(g) in saturated NaCl(aq). Of the possible ionic compounds that could precipitate from such a mixture (NaCl, NH_4Cl, $NaHCO_3$, and NH_4HCO_3), the least soluble is *sodium hydrogen carbonate* (sodium bicarbonate, $NaHCO_3$). It is isolated from solution by filtration and then converted to sodium carbonate (Na_2CO_3) by heating. According to this explanation,

3.2. Balance the reactions given below.

$$NaCl(aq) + CO_2(g) + NH_3(g) + H_2O(I) \rightarrow$$

$$2NaHCO_3(s) \xrightarrow{\Delta}$$

3.3. Using $CaCO_3$ (limestone), how can you produce the CO_2 gas you need to prepare $NaHCO_3$?

3.4. Write the Lewis structure of $CaCO_3$ with all resonances and show formal charges for each atom in the structure.

3.5. Describe the molecular geometry and propose a plausible hybridization scheme for the central atom in the ion CO_3^{2-}.

3.6. Compare the bond lengths of CO_3^{-2}, CO, and CO_2 in increasing order.

NaCl crystallizes in a face-centered cubic (fcc) structure. The density of NaCl is 2180 kg/m^3 and the ionic radius of Na^+ is 99 pm.

3.7. How many atoms are there in the unit cell? Which atoms occupy octahedral holes?

3.8. Calculate the length of the unit cell of NaCl and the ionic radius of the Cl^- ion (as pm).

3.9. The alkali metals react rapidly with oxygen to produce several different ionic oxides. Under appropriate conditions, generally by carefully controlling the supply of oxygen, the oxide M_2O can be prepared for each of the alkali metals. Lithium reacts with excess oxygen to give ……(A)…. and a small amount of ……(B)……. Sodium reacts with excess oxygen to give mostly ….(C)…… and a small amount of ……(D)……..Potassium, rubidium, and cesium react with excess oxygen to form …….(E)….., ……(F)….., and ……(G)…..

Fill in the blanks above (for A–G) with convenient formulas of metal oxides.

3.10. Draw the Lewis structures of oxide, peroxide, and superoxide ions.

3.11. Draw the molecule orbital energy level diagram of peroxide and superoxide ions and compare their bond lengths and energies.

When $LiClO_4$, $NaClO_4$, and $KClO_4$ crystallize from an aqueous solution that may or may not contain water molecules called water of crystallization as part of the solid structures, although no simple rule exists for predicting with certainty whether the ions will retain all or part of their hydration spheres in the solid state, cations with high charge densities tend to retain all or part of their hydration spheres in the solid state. When the cations have low charge densities, the cations tend to lose their hydration spheres; thus, they tend to form anhydrous salts. The ionic radius of Li^+, Na^+, and K^+ is 76 pm, 102 pm, and 138 pm, respectively.

3.12. Calculate the charge densities of the ions in C mm^{-3}.

3.13. Which perchlorate salt is most susceptible to form an anhydrous compound?